P-G-406D
October 27, 1977
SUPERSEDING
Fed. Spec. P-G-406C
September 23, 1975

FEDERAL SPECIFICATION

GLASS CLEANER, LIQUID
(CONCENTRATED AND READY-TO-USE)

This specification was approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal agencies.

1. SCOPE AND CLASSIFICATION

1.1 Scope. This specification covers commercial-type glass cleaners intended for use on all glass surfaces, plastics, ceramics, chrome, stainless steel, and painted surfaces (see 6.1).

1.2 Classification. The glass cleaners shall be of the following types and classes as specified herein (see 6.2).

Type I - Regular

Class 1 - Ready-to-use
Class 2 - Concentrated

Type II - Antifogging, ready-to-use

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issues in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

Federal Specifications:

PPP-B-636 - Boxes, Shipping, Fiberboard.
PPP-C-96 - Cans, Metal, 28 Gage and Lighter.
PPP-C-186 - Containers, Packaging and Packing for Drugs, Chemicals, and Pharmaceuticals.
PPP-G-460 - Glass Containers, One Gallon Capacity and Smaller for Other than Medicinal Products; Packaging and Packing Of.

Federal Standard:

Fed. Std. No. 123 - Marking for Shipment (Civil Agencies).

(Activities outside the Federal Government may obtain copies of Federal Specifications, Standards, and Handbooks as outlined under General Information in the Index of Federal Specifications and Standards and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402.

(Single copies of this specification and other Federal Specifications required by activities outside the Federal Government for bidding purposes are available without charge from Business Service Centers at the General Services Administration Regional Offices in Boston, New York, Philadelphia, Washington, DC, Atlanta, Chicago, Kansas City, MO, Fort Worth, Houston, Denver, San Francisco, Los Angeles, and Seattle, WA.

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FSC 7930

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Military Standards:

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129 - Marking for Shipment and Storage.

(Copies of Military Specifications and Standards required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

Laws and Regulations:

16 CFR 1500 - Federal Hazardous Substances Act Regulations.
49 CFR 170-189- Hazardous Materials Regulations.

(The Code of Federal Regulations (CFR) and the Federal Register (FR) are for sale on a subscription basis by the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402. When indicated, reprints of certain regulations may be obtained from the Federal agency responsible for issuance thereof).

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless a specific issue is identified, the issue in effect on date of invitation for bids or request for proposal shall apply.

American Society for Testing and Materials (ASTM) Standards:

D 56 - Flash Point by Tag Closed Tester.
D 3069 - Delivery Rate of Aerosol Products.
D 3074 - Pressure in Metal Containers.

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

National Motor Freight Traffic Association, Inc., Agent:

National Motor Freight Classification.

(Application for copies should be addressed to the American Trucking Associations, Inc., Traffic Department, 1616 P Street, N.W., Washington, DC 20036.)

Uniform Classification Committee, Agent:

Uniform Freight Classification.

(Application for copies should be addressed to the Uniform Classification Committee, Room 1106, 222 South Riverside Plaza, Chicago, IL 60606.)

3. REQUIREMENTS

3.1 Material. The manufacturer is permitted discretion in the formulation of commercial glass cleaners provided that type II, glass cleaner shall contain an antifogging agent. The manufacturer's recommended use dilution of the glass cleaner liquid concentrate and the ready-to-use glass cleaner (aerosol or finger-pump spray) shall meet the specified requirements in section 3.

3.2 Toxicity. The product shall contain no methanol when tested as specified in 4.3.10.

3.3 Flash point. When tested as specified in 4.3.1, the liquid glass cleaner shall have a flash point of not less than 120°F (48.8°C). Aerosol glass cleaner is excluded.

3.4 pH. The pH of the glass cleaner shall be 6.5 to 11.0, when tested as specified in 4.3.2.

3.5 Corrosion of aluminum alloy. The glass cleaners shall show no corrosive effect on aluminum alloys when tested as specified in 4.3.3.

3.6 Cleaning performance. When the glass cleaner is applied to glass surfaces and polished, it shall leave the surface free of dust, grime, ordinary soiling materials, streaks, and smears when tested as specified in 4.3.4.

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3.7 Antifogging, type II. The glass cleaner shall produce an antifogging action on glass when tested as specified in 4.3.5.

3.8 Aerosol dispenser requirements. The contents shall not react with nor be reacted upon by the interior surface of the dispenser or any of the dispenser components.

3.8.1 Propellant. The propellant shall be a hydrocarbon or a mixture of hydrocarbons. The use of fluorocarbons as the aerosol propellant gas is prohibited. The propellant shall be analyzed for fluorocarbons and vinyl chloride as specified in 4.3.9.

3.8.2 Container. The container shall be a commercial-type metal aerosol container of such construction as to assure acceptance of the finished package by common carriers operating in interstate commerce. The capacity of the container shall be sufficient to contain the specified net contents of the product with outage in accordance with commercial practice. The container shall not be overcoated to obliterate the original label.

3.8.3 Dispensing valve. The container shall have a continuous-spray dispensing valve with the performance characteristics specified in table I. The valve stem shall be fitted with a side discharge plastic actuator. The actuator orifice shall be round and free of all obstructions which will affect the flow and discharge pattern of the contents (see 4.2.2.3 and 4.3.6). The dispensing valve shall be designed so that when activated, none of the aerosol spray shall impinge on any part of the container. On operating the dispensing valve, the contents shall be ejected only through the orifice (see 4.3.6).

TABLE I. Valve performance characteristics

Characteristic	Requirement
Delivery rate at 21°C(70°F)	1.0 to 3.5 grams per second (g.p.s)
Leakage in 48 hours, maximum	3.0 milliliters (ml)

3.8.4 Cover cap. The dispensing valve shall be protected from accidental functioning and damage by a press-fitting cap of metal or sturdy plastic.

3.8.5 Unit container pressure stability. The filled unit containers shall show no evidence of leakage, distortion or other pressure-related defects when placed in a water bath at 55° ± 1°C (130° ± 2°F) for a period of 1 hour (see 4.3.7).

3.8.6 Laws and regulations (aerosol unit container). The product and container shall conform to the Federal Hazardous Substances Act and Department of Transportation Hazardous Materials Regulations.

3.9 Directions for use. The manufacturer's "Directions For Use" and precautions shall be clearly shown on the label of each container.

3.12 Sprayer units for non-aerosol glass cleaner. Each bottle of liquid glass cleaner specified in 5.1.1.1.1 and 5.1.1.1.2 shall be equipped with a leak-proof and functional commercial sprayer unit conforming to Bakan Plastics Model No. 5 or equal.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government.

4.2 Quality conformance inspection.

4.2.1 Sampling for inspection and acceptance. Sampling for inspection and acceptance shall be performed in accordance with the provisions set forth in MIL-STD-105, except where otherwise indicated herein.

4.2.2 Examination and test of the end item. The end item shall be examined in accordance with the classification of defects, inspection levels and acceptable quality levels (AQL), set forth below. The lot size, for purpose of determining the sample size in accordance with MIL-STD-105, shall be expressed in units of filled primary containers of glass cleaner of the same type contained herein.

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4.2.2.1 Examination of the end item aerosol unit container for defects. The sample unit shall be one filled unit container. Sampling and acceptance shall be in accordance with MIL-STD-105, inspection level I, AQL 2.5 expressed in terms of percent defective.

Examine	Defects
Unit containers	Corroded, bent, distorted, or leaking. Not useable. Not as specified.
Valve	Not as specified.
Plastic actuator	Not as specified.
Cover cap	Not as specified.

4.2.2.2 Examination of the end item container for defects in appearance of liquid glass cleaner. The sample unit for this examination shall be one filled primary container. The inspection level shall be S-2 with an AQL of 2.5 expressed in terms of percent defective.

Examine	Defects
Appearance of glass cleaner	Not a liquid. Not free flowing. Presence of dirt or foreign matter. Not homogeneous. Any evidence of segregation. Not clear. Not pourable, too viscous.

4.2.2.3 Examination of preparation for delivery. An examination shall be made to determine compliance with the requirements of section 5. Sampling shall be in accordance with MIL-STD-105. The sample unit shall be one shipping container fully prepared for delivery. The inspection level shall be S-2 with an AQL of 4.0 expressed in terms of percent defective.

Examine	Defects
Containers	Not as specified.
Contents	Not as specified.
Markings	Omitted; incorrect; illegible: improper size, location, sequence, or method of application.
Materials	Component missing or damaged.
Workmanship	Bulging or distortion of containers. Cushioning inadequate, improper or missing. nadequate closure.

4.2.2.4 Examination for closure, waterproofing, and banding of shipping container. When shipping containers are required to comply with PPP-B-636, examination for defects in closure, waterproofing, and banding shall be made in accordance with the appendix to that specification.

4.2.2.5 Testing the end item. The end items shall be tested for the characteristics listed in table II for each lot presented for examination for each type of material. Two aerosol unit containers of the glass cleaner shall be drawn from each of six separate shipping containers selected throughout the lot presented for test. The test sample shall consist of not less than 12 unit containers. Two 1/2-pints of liquid glass cleaner shall be drawn from each of six separate shipping containers, selected at random throughout the lot presented for test, and composited. The test sample shall consist of a 4-pint composite of liquid glass cleaner.

4.2.2.5.1 Unless otherwise specified, duplicate determinations shall be performed on the composite with no evidence of failure to meet the specified requirements.

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TABLE II. Instructions for testing the end item

Characteristics	Specification reference		Results reported as	
	Require-ment	Test method	Pass or fail 1/	Numerically to nearest 2/
Toxicity	3.2	4.3.10	X	
Flash point	3.3	4.3.1	X	1.0°C
pH value	3.4	4.3.2		0.1 pH unit
Corrosion of aluminum alloy	3.5	4.3.3	X	
Cleaning performance	3.6	4.3.4	X	
Antifogging, type II	3.7	4.3.5	X	
Aerosol dispenser:				
Aerosol spray impingement on container.	3.8.3	4.3.6	X	
Discharge of material other than through orifice.	3.8.3	4.3.6	X	
Delivery rate	Table I	4.3.6		0.5 ml
Leakage	Table I	4.3.7	X	
Filled unit container pressure stability	3.8.5	4.3.8	X	
Propellant gas for fluorocarbons	3.8.1	4.3.9	X	

1/ If failure is indicated, report either description of failure or numerical point of failure, as applicable.
2/ Test report shall include all values on which results are based.

4.3 Test procedures.

4.3.1 Flash point.

4.3.1.1 Liquid glass cleaner. Determine the flash point by the Tag Closed Cup Method in accordance with ASTM D 56.

4.3.2 pH value. Determine the pH with a pH meter using standard buffer solutions of pH 10 and pH 7 to standardize the pH meter.

4.3.3 Effect on aluminum alloy. Polish half of the area of one side of two 3 x 6-inch (7.8 x 15.6-cm) panels of aluminum alloy 3003. Place several drops of the solution or spray under test on the polished surface of one of the panels and cover with a watch glass. Treat the other panel in a similar manner using the control solution specified in 4.3.4.3. At the end of 6 hours at room temperature (20°C), remove the watch glasses, rinse the panels with water, and dry. The panels shall then be examined for corrosion and discoloration. The sample solution shall give no evidence of corrosion or discoloration in excess of that caused by the control solution.

4.3.4 Cleaning performance.

4.3.4.1 Equipment and reagents.

a. Gardner Heavy-Duty Wear Tester, or straight line "Washability" Machine, with 7-1/2 x 9-1/2-cm (2-9/10 x 3-7/10-inch) carrier.
b. Glass plates. The (0.31 cm) 1/8-inch glass should be cut to 19.1 x 30.5 cm (7-1/2 x 12 inch).
c. Cheesecloth, grade 40. his should be cut into about 16 inch (41 cm) square, so that each piece weighs about 13 g.
d. Wood, 6.5 x 9.0 x 2.6 cm (2-1/2 x 3-1/2 x 1 inch).
e. Soils.

Mineral oil, with a viscosity of 300-320 SUS at 38°C (100°F). Kaopaque - 10 clay, available from Georgia Kaolin Co. Spangler's synthetic sebum, whose composition is:

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	Weight percent
Palmitic acid	10
Stearic acid	5
Coconut oil	15
Paraffin wax	10
Synthetic Spermaceti	15
Olive oil	20
Squalene	5
Cholesterol	5
Oleic acid	10
Linoleic acid	5

Perchloroethylene, as a carrier for the soils. Technical grade chemicals may be used in Spangler's synthetic sebum.

f. Chemicals.

Mineral spirits (naphtha, V.M. & P.)
Dipropylene glycol monomethyl ether (DOWANOL DPM-Dow Chemical Company or equal)
Isopropanol
Pluronic F 108 surfactant (Wyandotte Chemicals Corporation) or equal
Ammonium hydroxide (28% in water)

g. Preval Spraymaker No. 468, and glass container No. 469 for applying soils and window cleaners. Available from Precision Valve Corporation.

h. Masking tape, 1.3 cm (1/2 inch) width.

4.3.4.2 Soil formulations (by weight percent)

	(A) Exterior	(B) Interior
Mineral oil	0.75	0.5
Clay	0.75	0.5
Synthetic sebum	-	0.5
Perchloroethylene	98.5	98.5

The clay should be sprinkled in last to prevent or minimize agglomeration.

4.3.4.3 Control glass cleaner formulation (weight percent).

Isopropanol	4.00
Distilled water	90.90
Dipropylene glycol monomethyl ether	4.00
Pluronic F 108	0.10
Ammonium hydroxide (28% in water)	1.00

4.3.4.4 Preparation of glass plates. Thoroughly clean the glass plates before and after use with cheesecloth in solvent baths in the order listed below:

a. Mineral spirits
b. Dipropylene glycol monomethyl ether
c. Isopropanol
d. Water

After the water rinse, dry the glass plates with cheesecloth. To insure complete cleanliness of the plates, expose them to live steam. A white, highly reflective appearance is taken on by the glass in those areas not thoroughly clean. In such cases, the glass plates should be exposed to the entire solvent cleaning procedure again.

4.3.4.5 Application of soils. Apply the soils uniformly, using the Preval Spraymaker, with 5.5 $\pm$ 0.1 g being applied to each clean glass plate. This calculates to about 0.0825 g of actual soil per plate. Uniform application of the soils is accomplished by using a fairly slow sweeping motion of the Preval Unit over the glass.

4.3.4.6 Aging of soils. Age the soiled glass plates at 50°C for two hours, followed by cooling to ambient temperature, prior to application of the cleaners.

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4.3.4.7 Cleaning the glass plates.

a. Place masking tape (1/2 inch) over the center of the soiled glass plate, leaving two 8.9 x 30.5-cm (3-1/2 x 12-inch) soiled areas exposed.
b. Drape one side of the soiled glass plate with cheesecloth to avoid overspray when applying glass cleaner to the remaining exposed area.
c. Apply 2 g of glass cleaner uniformly over the exposed glass plate area using the Preval Spraymaker dispenser or aerosol can.
d. The glass cleaner is allowed to penetrate the soiled area for one minute. Meanwhile, the cheesecloth drape is removed.
e. The treated portion of the glass plate is centered and placed under the Gardner apparatus equipped with the 7-1/2 x 9-1/2 cm- carrier. A 13-g piece of cheesecloth should be wrapped around the wood and inserted snugly into the carrier. The cheesecloth should be wrapped in such a way that its ends are along the side of the wood block, not along the cleaning area. Weights should be added to or taken from the carrier so that the total weight of carrier, wood, and cheesecloth is about 454 g.

NOTE: The trays normally used for hard surface cleaning tests are not needed. Therefore, 1/8-inch vinyl tiles are required to keep the glass plates rigid during the cleaning cycles.

f. Run Gardner for ten cycles (20 strokes), turn over cheesecloth in holder, and continue wiping for an additional ten cycles (20 strokes). Twenty cycles total.
g. Repeat Steps b through f using a second glass cleaner on the remaining soiled portion (8.9 x 30.5 cm) of the same glass plate.

4.3.4.8 Rating of cleaners. Both sides of the cleaned glass plates are rated for cleaning ability, streaking, and smearing tendency according to the following scales.

a. Cleaning.

5 = Excellent soil removal.
4 = Good soil removal.
3 = Moderate soil removal.
2 = Fair soil removal.
1 = Poor soil removal.

b. Streaking.

4 = None.
3 = Slight streaking.
2 = Moderate streaking.
1 = Severe streaking.

c. Smearing, when a figure 8 is drawn with a tissue wrapped around a finger.

4 = None.
3 = Slight smear.
2 = Moderate smear.
1 = Severe smear.

The performance of the liquid glass cleaners shall be equal to the performance of the control glass cleaner (see 4.3.4.3) on the cleaning, streaking, and smearing scales. Three replicate tests should be used to evaluate each glass cleaner.

4.3.5 Antifogging, type II.

4.3.5.1 Control antifogging glass cleaner formulation (weight percent).

Isopropanol	5.00
Dipropylene glycol monomethyl ether	3.00
UCAR silicone surfactant L-7001 or L-7604 (Union Carbide Corp.)	1.00
Ammonia (28%)	0.25
Water	90.75

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4.3.5.2 Preparation of glass plates. Make 6 x 12 inch-(15.4 x 30.8 cm) mirror test plates by splitting a standard 12-inch-square mirror. Tape the edges with masking tape. Thoroughly clean the mirror test plates before and after use with cheesecloth in solvent baths in the order listed below:

a. Mineral spirits (naphtha, V.M.&P.)
b. Dipropylene glycol monomethyl ether
c. Isopropanol
d. Water

After the water rinse, dry the glass plates with cheesecloth. To insure complete cleanliness of the plates, expose them to live steam. A white highly reflective appearance is taken on by the glass in those areas not thoroughly clean. (In such cases, the glass plates should be exposed to the entire solvent cleaning procedure again). Drape one-half of a test mirror plate with a towel and apply both the control (see 4.3.5.1) and the sample defogging glass cleaners to the other half of the same mirror in a small area leaving a portion untreated for comparison. Wipe until the surface is just damp and then let dry. Hold the mirror 6 inches above a steam source and move the mirror back and forth in order to get even distribution of the steam. Observe the steam condensation as compared to the untreated side. Sometimes it is helpful to cool the mirror in a refrigerator prior to steaming. When the mirror is cool, conduct a defogging test by simply blowing one's breath across the mirror. The antifogging action of the sample glass cleaner shall be equal to or better than the antifogging action of the control glass cleaner (see 4.3.5.1).

4.3.6 Delivery rate. Determine the delivery rate at 21°C (70°F) of the filled aerosol container in accordance with ASTM D 3069. Examine the container for aerosol spray impingement on any part of the container and discharge of material other than through the actuator orifice.

4.3.7 Leakage. Remove the cover cap from the unit container. Submerge the container in a water bath that is deep enough to allow the container to stand upright and be completely covered with water. Fill a wide-mouth graduated cylinder (graduated in milliliters) with water, invert it, and place it over the container so that it does not touch the valve and actuator. Maintain the temperature of the water bath at 21°C (70°F). Allow the container and graduate to stand for 48 hours.

4.3.8 Unit container pressure stability. Submerge filled unit container in an automatically controlled, circulating, constant temperature bath at 23° ± 3°C (73.4° ± 6°F). Place a safety shield in front of the bath in case the container ruptures. Set the bath at 55° ± 1°C (131° ± 2°F) and allow the temperature to rise. Place the container in the bath for a period of 1 hour. Using gloves and face shield, remove the container from the bath and examine for leakage, distortion, or other pressure related defects.

4.3.9 Analysis of propellant gas for fluorocarbons and vinyl chloride.

4.3.9.1 Special apparatus and materials. The chemical analysis procedure described herein requires the following apparatus and materials:

A can piercing, pressure measuring device recommended in the ASTM D 3074 for determining the internal pressure of aerosol products. The device is available as Model CBA-6 from Builders Sheet Metal Works, Inc., 110 Wooster Street, New York, New York 10012.

A temperature programmable gas chromatograph (Hewlett-Packard Model 5750 gas chromatograph, or equivalent) equipped with a 0.5-ml gas sampling valve and thermal conductivity detector.

Two stainless steel gas chromatography columns, 6 feet long by 1/8" outside diameter, packed with 60-200 mesh silica gel.

Standard propellant gases in lecture bottles from Matheson Gas Products (or another supplier): vinyl chloride, fluorocarbon Nos. 11, 12, 114, 115.

Matheson lecture bottle control valve Model 58, or equivalent.

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4.3.9.2 Gas chromatographic analysis. Condition the chromatographic columns at 200°C with normal helium as the carrier gas at 75-100 ml/minute for one hour. Remove the actuator and cap of an unused aerosol container. Puncture the can using the pressure measuring device and connect the device to the gas chromatograph. Triplicate chromatographic analyses shall be done on the aerosol propellant sample and on mixtures of the aerosol propellant with each of the five standard propellant gases. For mixing the standard propellants with the aerosol propellant sample, use a "T" connection (Swagelock or equivalent). The "T" connection shall be removed entirely when the aerosol propellant sample alone is to be analysed. Bubble each gas sample through the sampling valve at a moderate rate for two minutes before each run is made to purge the valve of any other gases. Analyze the samples using a thermal/conductivity detector. The temperature program shall be 100°C to 200°C at 6°C per minute with a 2-minute post injection hold at 100°C and a 5-minute upper limit hold at 200°C. Helium is used as the carrier gas at the rate of 75-100 ml per minute. There shall be no detectable material in the aerosol propellant sample having the same retention time as any of the five standard propellant gases.

4.3.10 Detection of methanol in glass cleaner.

4.3.10.1 Special apparatus. A temperature programmable gas chromatograph (Hewlett-Packard Model 5754 or equivalent) equipped with dual flame ionization detectors and two stainless steel gas chromatographic columns, 6 feet long by 1/8-inch outside diameter, packed with Poropak Q-S.

4.3.10.2 Procedure. Condition the chromotographic columns at 200°C with helium as the carrier gas at 75-100 ml/minute for one hour. Set the injection port and detectors at 250°C. Inject 3 ul of sample and program the temperature from 100°C to 200°C at 8°C/min with a 2-minute post injection hold at 100°C and a 5-minute upper limit hold at 200°C. Inject 3 ul of 0.5 percent (by volume) agueous methanol and follow same procedure as with the sample. There shall be no detectable methanol in the glass cleaner having the same retention time as the agueous methanol.

5. PREPARATION FOR DELIVERY

5.1 Packaging. Packaging shall be level A, or commercial as specified (see 6.2).

5.1.1 Level A.

5.1.1.1 The eight-ounce quantity of one type of glass cleaner as specified (see 6.2) shall be packaged in a container conforming to 5.1.1.1.1, 5.1.1.1.2, or 5.1.1.1.3. The one-gallon or five-gallon quantity shall be packaged in a container conforming to 5.1.1.1.2 or 5.1.1.1.3. The sprayer units for the eight-ounce bottles (see 3.12) shall be secured to the bottle.

5.1.1.1.1 Glass bottle. Glass bottle shall be of good commercial quality and free from defects such as cords, stones, blisters, or thin spots which may impair the strength or serviceability. The glass bottle shall be furnished with a screw cap and with an inner seal. Each bottle shall be further secured with a "Seal A", closure conforming to PPP-C-186.

5.1.1.1.2 Plastic bottle. Plastic bottle shall be of high density polyethylene of the type normally used for this item. The plastic bottle shall be furnished with a screw cap and an inner seal. Each bottle shall be further secured with a "Seal A" closure conforming to PPP-C-186.

5.1.1.1.3 Metal can. Metal can provided with screw cap shall conform to PPP-C-96, type V, class 4 or type VIII. The screw cap shall be of metal or plastic provided with inner seal and self-shrinking outer-seal.

5.1.1.2 Aerosol dispenser. The aerosol dispenser described in 3.8 does not require packaging.

5.1.2 Commercial. The glass cleaner shall be packaged in containers normally used in commercial practice. The complete package shall be designed to protect the item against damage during shipment, handling and storage.

5.2 Packing. Packing shall be level A, or commercial as specified (see 6.2).

5.2.1 Level A.

5.2.1.1 Glass bottles. The filled glass bottles shall be packed in accordance with PPP-G-460 for level A packing.

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5.2.1.2 Metal cans. The filled metal cans shall be packed in accordance with PPP-C-96 for level A packing.

5.2.1.3 Plastic bottles. The filled plastic bottles shall be packed in a box conforming to PPP-B-636, class weather resistant. The box shall be fitted with dividers, partitions, or separators fabricated of fiberboard of the same description as the box. The box shall be 1/8 to 3/8 inch higher than the bottles. The box shall be closed, waterproofed and banded in accordance with the appendix to the box specification.

5.2.1.4 Aerosol cans. Twelve aerosol cans shall be packed in a box conforming to PPP-B-636, class weather resistant. The box shall be closed, waterproofed, and banded in accordance with the appendix to the box specification.

5.2.2 Commercial. The glass cleaner shall be packed to insure safe delivery at destination, to provide for safe redistribution by the initial receiving activity, and shall be acceptable by common carrier under National Motor Freight Classification or Uniform Freight Classification.

5.3 Marking. The unit and shipping containers and unitized loads (when applicable) shall be marked in accordance with Fed. Std. No. 123 for civil agencies or MIL-STD-129 for military agencies, as specified (see 6.2).

5.4 Unitization. When shipments to government depots are full car or truck load the shipping containers shall be unitized for shipment in accordance with normal commercial practice. The unitized load shall not exceed 2,500 pounds in weight, 63 inches in height, 56 inches in length, and 45 inches in width.

6. NOTES

6.1 Intended use. The types I and II glass cleaners are intended for the cleaning and polishing of all glass surfaces, plastics, ceramics, chrome, painted surfaces, and stainless steel and are capable of removing dust, grime and other matter commonly soiling such surfaces. In addition, the type II glass cleaner is intended for glass surfaces where antifogging properties are required and should exhibit these properties for several hours.

6.2 Ordering data. Purchasers should select the preferred options permitted herein, and include the following information in procurement documents:

(a) Title, number, and date of this specification.
(b) Type and class required (see 1.2).
(c) Packaging and packing required (see 5.1 and 5.2).
(d) Marking required (see 5.3).
(e) When sprayers are required (see 3.10).

6.3 New classification changes. The following classification changes have been made:

Old designation	New designation
Type I - Regular	Type I - Regular
	Class 1 - Ready-to-use Class 2 - Concentrated

MILITARY INTEREST:

The Department of Defense has waived coordination of this document until further notice.

Preparing activity:

GSA-FSS

CIVIL AGENCY COORDINATING ACTIVITIES:

GSA-FSS
VA-DMA
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Orders for this publication are to be placed with General Services Administration, acting as an agent for the Superintendent of Documents. See Section 2 of this specification to obtain extra copies and other documents referenced herein. Price 70cents each.

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